Kathmandu University

Department of Computer Science and Engineering

Dhulikhel, Kavre

A Project Report on

"Sirani - Your Therapy Buddy"

[Code No: ENGG 102]

(For partial fulfillment of 2nd Semester in Computer Science)

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Bona fide Certificate

This project work on

“SIRANI - YOUR THERAPY BUDDY”

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who carried out the project work under my supervision.

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Date: 4th May 2022

Abstract

The COVID-19 pandemic has negatively impacted many lives, especially those of students. This project, Sirani, focuses on the mental health and well-being of the user to help them effectively manage their mental health. It includes a personal goal setting feature, sleep and mood tracking, and a mental health resources section. As required, this project is developed using the programming language C++ along with the Qt framework to design a user-friendly GUI. SQLite is used as the database management system. Sirani intends to make mental health accessible, comfortable, and non-stigmatizing.

Keywords: C++, Qt framework, SQlite

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Acronyms/Abbreviations

WHO	World Health Organization
ESA	Emotional Self Awareness
GP	General Practitioner
OS	Operating Software
PC	Personal Computer
MHA	Mental Health America

Chapter 1: Introduction

1.1 Background

Although our lives have been drastically altered over the past 2 years, people are returning to their normal lives following the end of lockdown. The mental health of people has taken a heavy hit throughout the lockdown and re-adapting back to our lives is equally challenging. Amidst all this, students have a more difficult time focusing on their studies, especially with the never-ending fear of another wave of Corona. The World Health Organization (WHO) has identified mental health as an integral component in COVID-19 response. [1] Loneliness and substance abuse have been a common problem among youths and adults alike. “Addressing mental health and substance use problems in young adults, both during and after the COVID-19 pandemic, is an imperative,” says Viviana Horigian from the University of Miami in Coral Gables, Florida. [2]

The issue of mental health is not unique to the pandemic but has always been of prime importance. “There is evidence that self-monitoring, on its own, is a therapeutic activity via increasing self-awareness”. [3] Hence, A relatively new approach to mental health is the use of software to monitor mood, stress, coping strategies, activities, eating, sleeping, and exercise. A trial conducted with 163 participants regarding a mobile type application found out that the data collected from the application can be reviewed by a doctor which assisted the communication between the doctor and the patient and helped the doctor understand the patient better. It was concluded that "Monitoring mental health symptoms appears to increase emotional self-awareness (ESA) and implementing a mental health program in primary care and providing frequent reminders, clinical resources, and support to general practitioners (GP) substantially improved mental health outcomes for the sample as a whole.” [4]

Based on this knowledge, we have developed “Sirani - Your Therapy Buddy” designed to help users more effectively manage their mental health. It comprises a multitude of features ranging from mood and sleep tracker to a collection of resources with an interactive UI designed with Qt framework and C++ to help our users feel better.

1.2 Objectives

- Keep track of the user’s daily, monthly and annual goals to help them achieve their short and long-term goals.
- Provide a graphical representation of the user’s sleep and mood patterns to help them keep track of their past and current state.
- Provide a collection of carefully chosen resources regarding mental health and ways to uplift it.
- Prompt the user for a weekly test to increase their self-awareness.

1.3 Motivation and Significance

In a world, so demanding, it is not only important to take care of physical health but also of our mental health. But Nepal being a conservative society, people are not only hesitant to consult a psychiatrist about their mental health but it is often overlooked and stigmatized. Depression, Anxiety, Attention-deficit/hyperactivity disorder (ADHD), Obsessive-Compulsive Disorder (OCD) are some of the most common mental health illnesses. In a study conducted among 1647 participants aged 13 years and above in three districts of Nepal in 2019, 13.2% of adults and 11.2% of children were suffering from some sort of mental disorder. [5]

Hence, we have designed Sirani to help people not just to keep track of their mental health but also to encourage going to therapy if needed. This is done via a daily mood and sleep tracker, and carefully chosen resources to encourage visiting professionals who can help them for further professional counseling.

Chapter 2: Related Works/ Existing Works

Though mental health is crucial for us, there aren't many effective apps on the market. Most of the existing apps are either paid apps or have in-app purchases. Some of the apps that are beneficial for mental health are Moodfit, Happify, Calm, Moodmission, etc.

Moodfit [6] is a mental health app that primarily helps to reduce stress to build resilience. It provides us with a set of tools for our mental health by analyzing the nature of our mood. Using Moodfit, one can easily identify what affects their mood and feeling which may be their diet, exercise schedule, sleep cycle, medication, etc.

Pros:

- Record and analyze our mood ups and downs.
- Breathing exercises that provide relief in stressful situations.
- Article for awareness that helps to understand our moods and behavior.
- Customizable according to individuals' needs and goals.

Cons:

- It has in-app purchases. So extra money is needed for advanced features.
- It doesn't have the contact info of a professional.

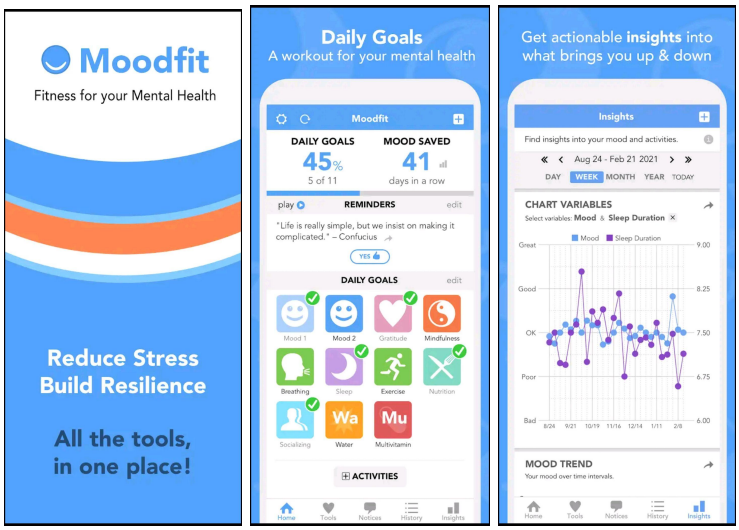

Fig 2.1: Moodfit App

Calm [7] is a free mental health app for meditation and sound sleep. It provides soothing nature sounds and short stories to motivate and eliminate distraction for stress-free focus in work.

Pros:

- Breathing exercises for immediate relief.
- Exclusive soothing sound for meditation, focus, and sleep.
- Calm masterclass by professional experts.
- Daily streaks for tracking progress.
- Sleep stories for sound sleep.

Cons:

- The majority of content needs a premium subscription.
- The app isn't intuitive.
- Audio is poorly optimized.

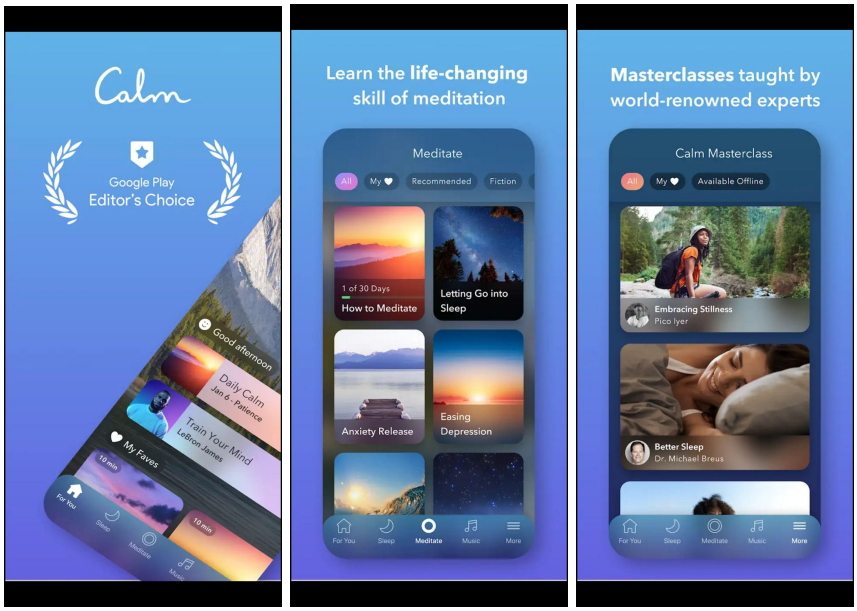

Fig 2.2: Calm App

Todoist [8] is a free mental health app that organizes tasks and sets a reminder for reducing the stress that refocuses their mind.

Pros:

- Organize tasks in order of their importance.
- Due dates and reminders for completing our task in time.
- Helps to refocus on one task at a time.
- Productivity trend for tracking progress.
- Color code to streamline the list to be done.

Cons:

- Most of the guided content is for Premium users only.
- Shared projects sometimes don't sync.

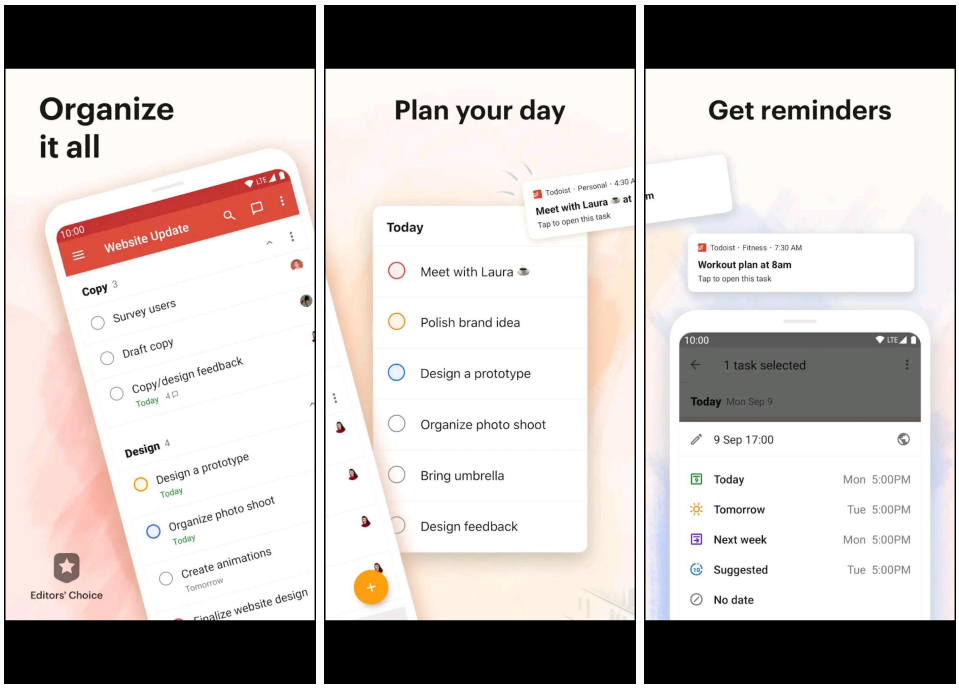

Fig 2.3: Todoist App

The current app market is saturated with mental health-related apps. It has become more of a business model than providing genuine help to needy people. Most apps provide limited features unless we're subscribed. Furthermore, they have a sophisticated UI that is hard to navigate for new users and very few apps exist that provide community-based resources.

There are many apps on the market that only perform a few tasks. Several apps track sleep and provide soothing music. Few apps aim to boost mood. Handful apps help to find professional help and related articles and videos. Several of them act as reminders and others help to track mental status. But these apps don't act as an all-in-one package for improving mental health. So we have designed Sirani such that it has all of the collective features in one to provide better mental health.

Chapter 3 : Design and Implementation

3.1 Discussion and Snippet of Project

Login

Upon running the project, the user is prompted to a login page, where they have to input their user credentials. SQLite is used to store username and password assigned during signup.

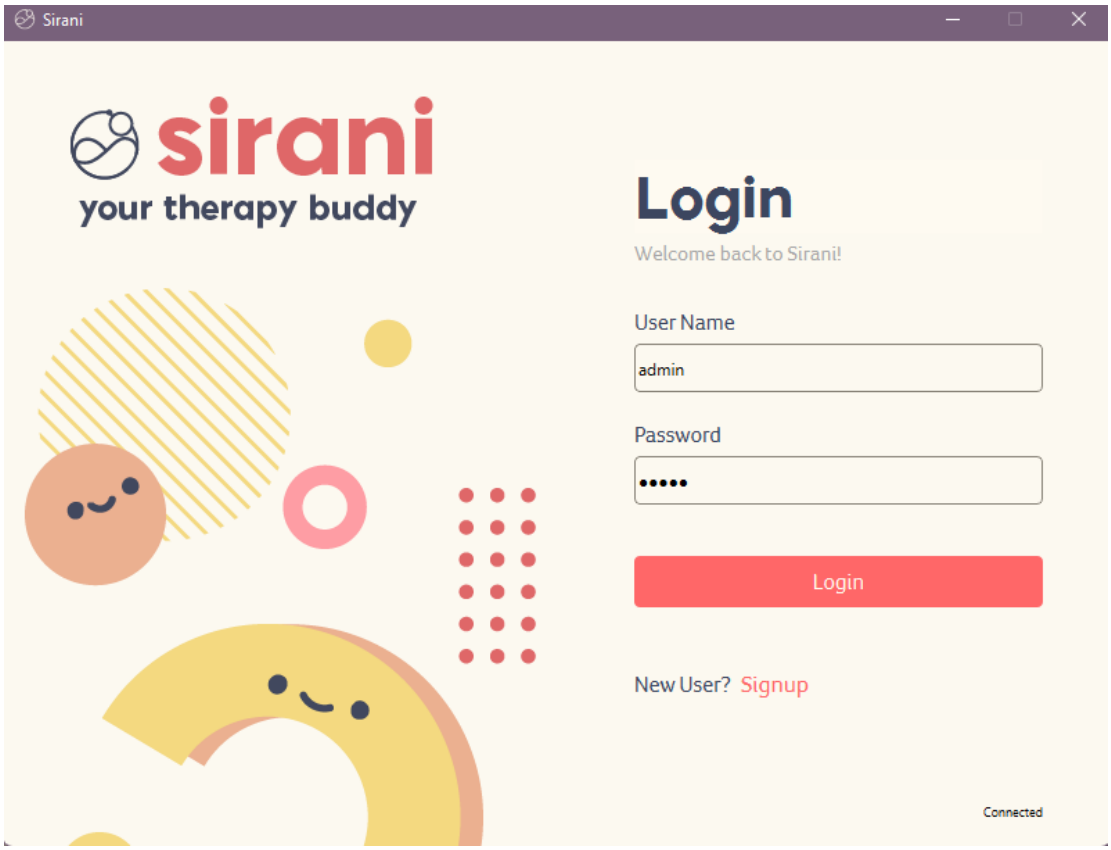

Fig 3.1.1: Login Page

Signup

On the login screen, clicking the signup button prompts users to a new signup page. For creating an account, the user needs to enter their full name, username, email, and password. A verification system is in place to ensure that email are entered correctly and password is strong. It also checks for duplicate usernames. The program works by creating a mood, sleep and goals table with their username as the name of the table for each user that signs up.

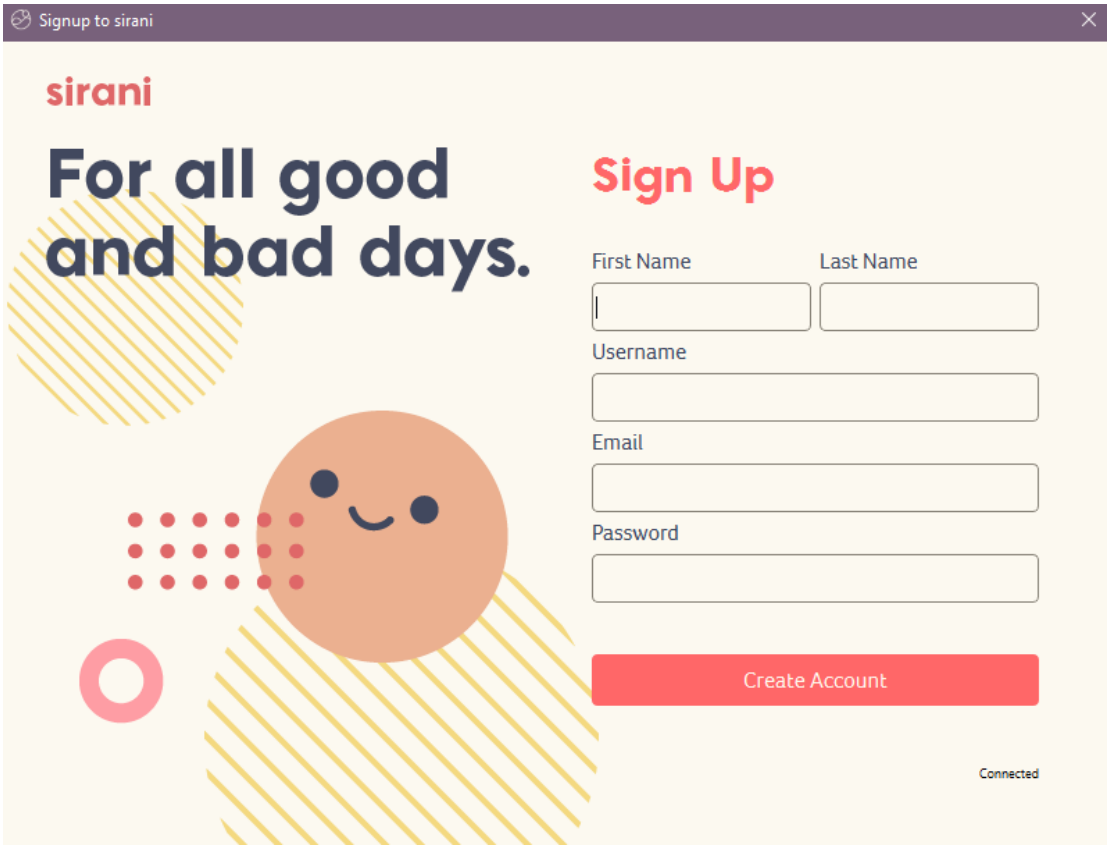

Fig 3.1.2: Sign Up Page

Dashboard and Navigation bar

The dashboard is the landing page that displays the user's sleep and mood patterns, along with a new inspirational quote every time you login. The nav bar is displayed in every page and consists of several buttons to navigate throughout the application. It also consists of a clock to show the date and time.

Fig 3.1.3: Dashboard

Goals

This feature allows users to set, view and remove daily, monthly, and yearly goals.

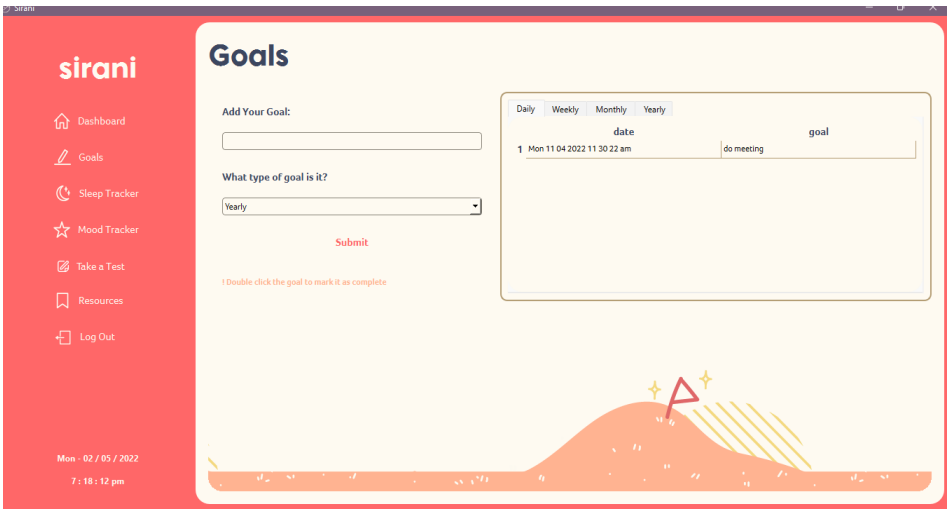

Fig 3.1.4: Goals

Sleep Tracker

In this section, users enter their daily sleeping hours, and the system displays a graphical representation of their sleeping patterns. In addition, users can erase all their data and start over.

Fig 3.1.5: Sleep Tracker

Mood Tracker

Users can enter their mood throughout the day and can view the frequency of each mood they've felt throughout the week in a graph and get suggestions based on how they're feeling. The user can choose from four different emotions: happy, sad, calm, and angry.

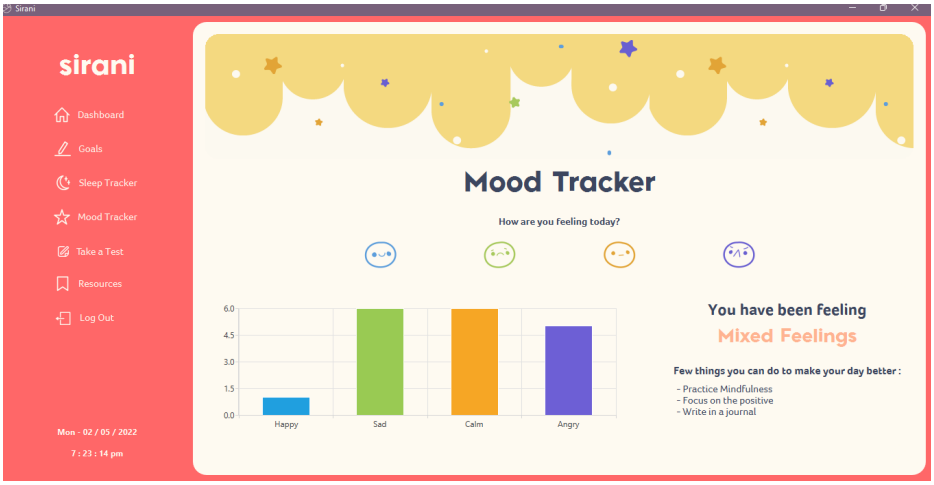

Fig 3.1.6: Mood Tracker

Resources

The resource section provides information on guidance and counseling cells in KU, helplines in Nepal, and a link for self-assessment.

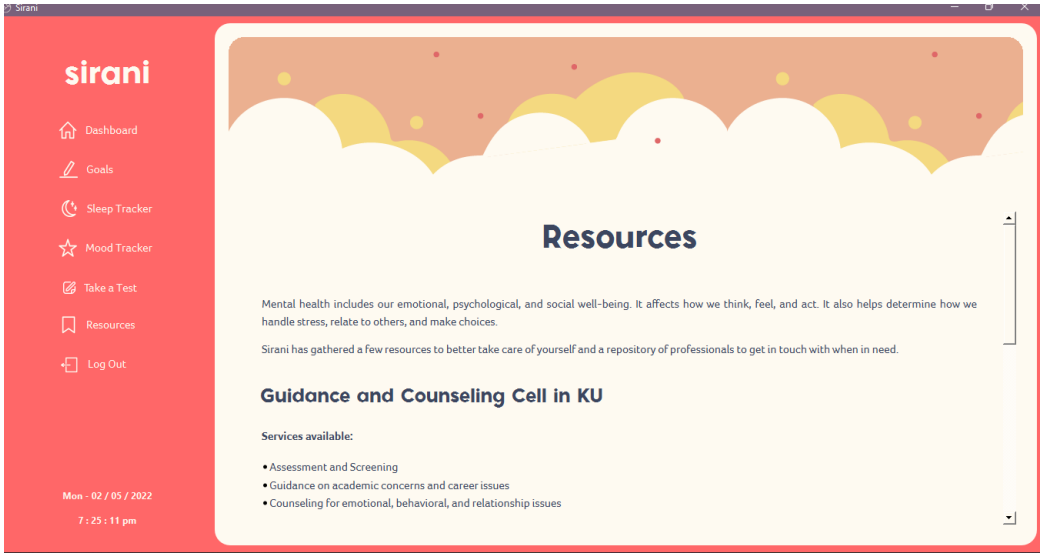

Fig 3.1.7: Resources

Take a test

Users can take a series of mental health tests to assess their mental health by clicking this button from the MHA website.

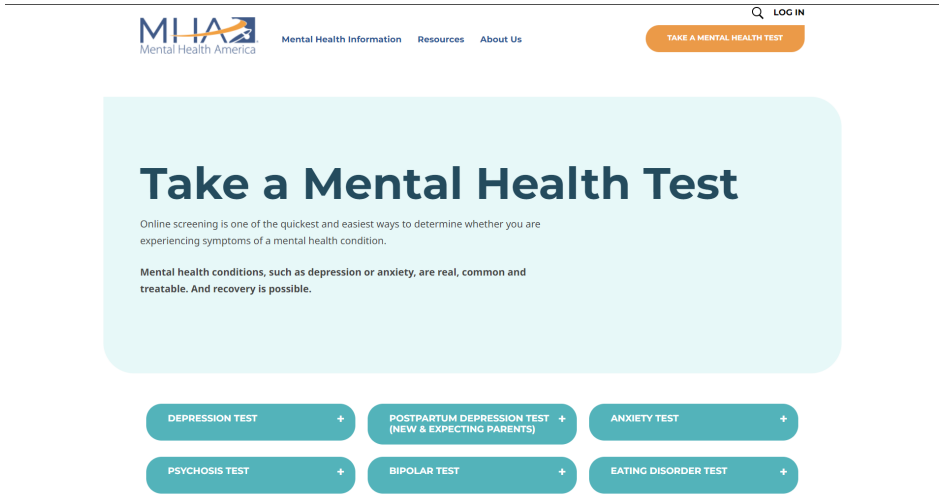

Fig 3.1.8: MHA Mental Health Test

Log Out

By clicking this button, the user can log out, but a dialog box asks for confirmation that they really want to log out.

3.2 UML diagrams

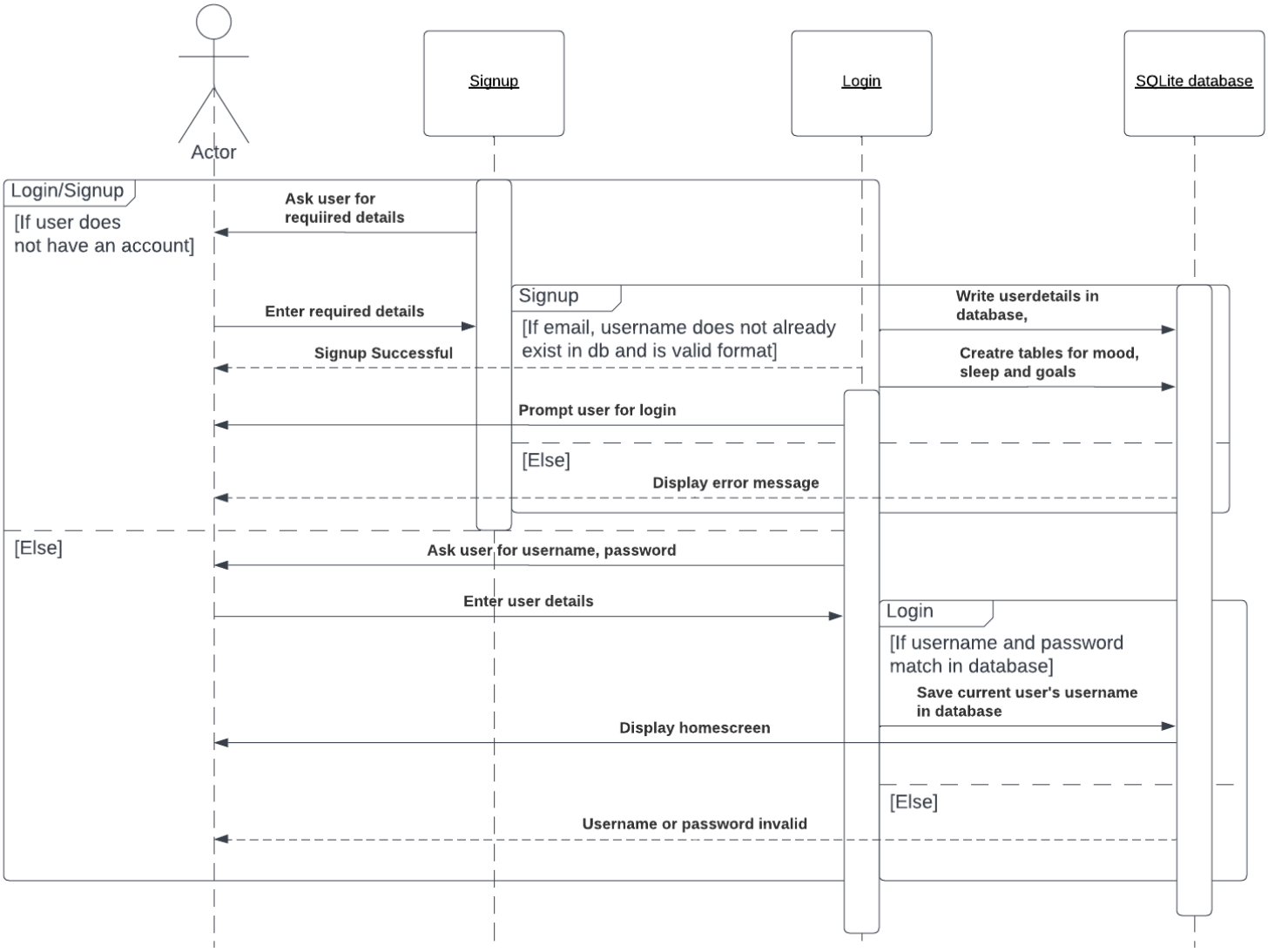

Fig 3.2.1: UML Sequence Diagram

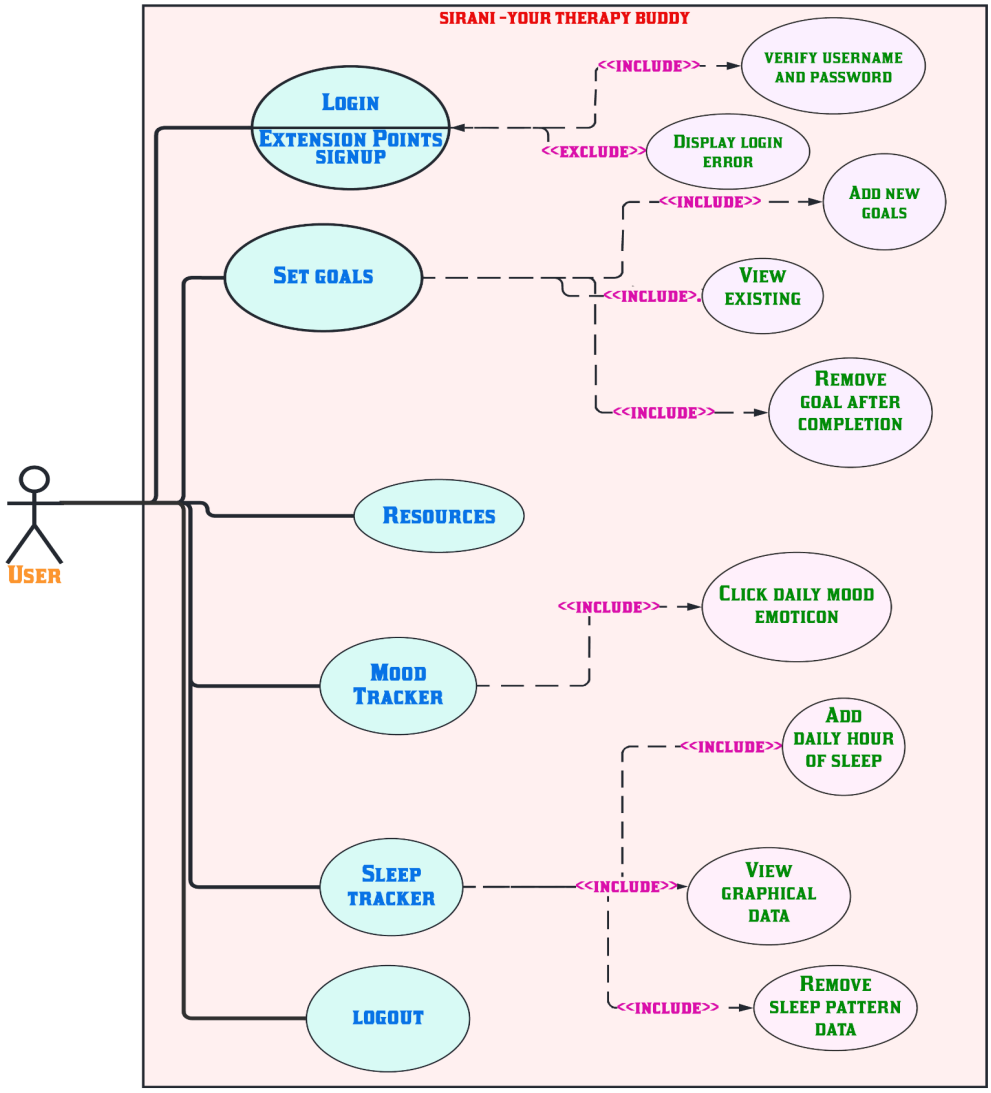

Fig 3.2.2 : UML Use Case Diagram

3.3 System Requirement Specification

3.3.1 Software Specifications

3.3.1.1 Front End Tools :

C++ with Qt framework

3.3.1.2 Back End Tools :

SQLite

3.3.2 Hardware Specifications

Any PC having Windows 7 or higher.

Chapter 4: Discussion of the achievement

4.1 Features

- Invalid email and password format detection when signing up.
- Digital clock that displays date and time in the main menu.
- Set and remove daily, monthly, and yearly goals.
- Graph for number of hours of sleep and frequency of each mood for the past 7 data.
- Daily mood record with average mood indicator for the week along with suggestions based on your mood.
- Daily sleep tracker with table to view all records.
- Resources containing professional contact details and link for self assessment.

4.2 Challenges

- Familiarizing with QT framework's widgets, layouts, and stylesheet.
- Learning git and github for version control and to collaborate among group members.
- Selection of color palette, icons, logo, backgrounds, and layouts for a responsive user interface.
- Integrating SQLite into the project for backend.
- Using QtCharts to display sleep and mood records of the last 7 days stored in the database.

4.3 Achievements

- Colorful and modern UI with user friendly interface and easy navigation.
- Working with a team on a small-scale project.
- Designing an app that can be used on a small scale for tracking mental health.

Chapter 5 : Conclusion

Sirani is a simple, user-friendly app that helps users keep track of their mood and sleep and provides them with resources for better coping. While there are a few limitations and future enhancements needed, Sirani does a good job of tracking and providing resources.

5.1 Limitation

- Use of sqlite over mysql limits our project to interact over a network.
- System requires at least data of 7 days for the graph demonstration of sleep and mood tracker.
- Mood can only be inserted once although there may be more than one mood in a day.
- Accomplished goals cannot be reviewed in future as there is no provision of saving completed /uncompleted goals for later.
- There is no deadline for the goals to track for better understanding on whether the goal is achieved on time or not.
- There is no real-time update of the graph when the user inputs a new data

5.2 Future Enhancement

Sirani aims to provide an unstigmatized and safe environment for the user's mental health. To make this more efficient, we can make a few improvements as follows:

- Addition of a notification system to remind users to input their data
- Addition of deadline concept in set goals section .
- Provision to save completed /uncompleted goals for later if the user wishes to.
- Use of a database which requires a server and interaction of client and server architecture over a network.
- Arrangement of adding mood conditions as much as the user wishes in a day.

References

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Appendix

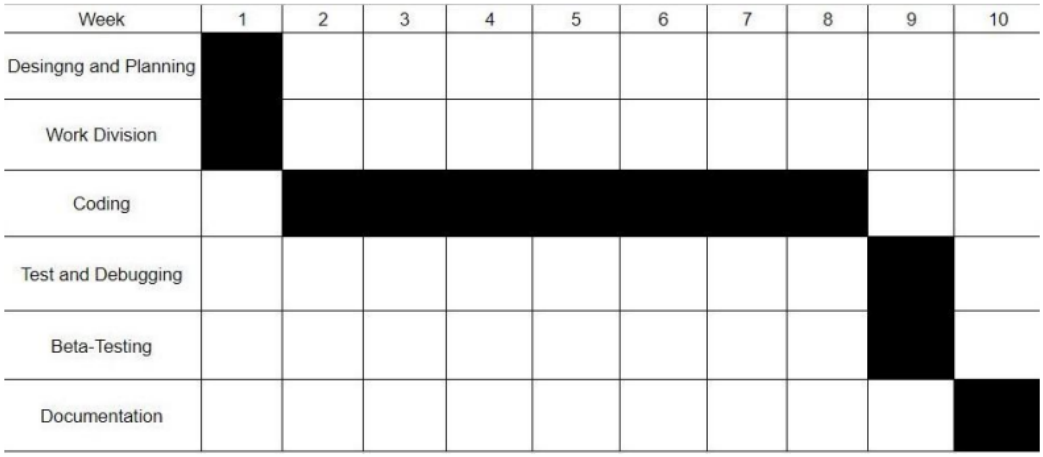

Fig A.1: Gantt Chart